

TONGDE, China

WMO: 529570

Lat: 35.267N

Lon: 100.650E

Elev: 10794

StdP: 9.80

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.2	(c) -10.8	(d) -35.2	(e) 1.1	(f) 3.4	(g) -30.5	(h) 1.5	(i) 3.9	(j) 31.1	(k) 29.3	(l) 23.1	(m) 27.8	(n) 8.6	(o) 90	(p) 0.737

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.2	(c) 71.4	(d) 52.1	(e) 68.6	(f) 50.8	(g) 66.0	(h) 49.5	(i) 56.0	(j) 65.8	(k) 54.3	(l) 63.4	(m) 52.6	(n) 61.0	(o) 6.1	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.9	89.8	59.7	51.1	84.2	57.7	49.5	79.2	56.4	29.1	66.2	27.7	63.0	26.4	61.4	62.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	17.8	14.4	-21.4	76.5	4.7	2.0	-24.7	77.9	-27.5	79.0	-30.1	80.2	-33.5	81.6
			-22.1	59.1	4.5	1.7	-25.3	60.3	-27.9	61.3	-30.4	62.2	-33.7	63.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	10.4	17.5	27.0	37.0	45.2	50.0	53.0	52.8	46.2	34.8	22.7	14.0		
	DBStd	15.93	7.04	6.56	6.65	5.75	5.29	4.66	4.11	4.36	5.52	5.65	5.70	5.28		
	HDD50	6051	1229	911	713	390	166	56	17	19	142	472	819	1117		
	HDD65	11208	1694	1331	1178	839	612	450	373	379	565	936	1269	1582		
	CDD50	319	0	0	0	2	19	56	110	105	27	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	18	0	0	0	1	0	1	2	10	4	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.8	7.2	7.3	8.2	7.9	7.6	6.5	5.8	5.7	5.9	6.0	6.8	7.2		
(15) Precipitation	PrecAvg	16.8	0.1	0.1	0.3	0.7	2.3	3.2	3.8	3.0	2.3	0.9	0.1	0.0		
	PrecMax	23.6	0.3	0.5	1.0	2.3	4.8	5.4	6.0	5.6	4.4	3.7	0.6	0.3		
	PrecMin	12.5	0.0	0.0	0.0	0.0	0.4	1.0	2.0	1.5	0.8	0.0	0.0	0.0		
	PrecStd	3.0	0.1	0.1	0.2	0.4	1.0	1.0	1.2	1.0	0.9	0.7	0.2	0.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.2	47.4	55.1	66.6	70.2	71.0	73.8	75.4	73.0	60.0	51.3	44.6		
		MCWB	26.9	29.3	33.6	40.0	44.9	49.2	53.7	54.0	55.0	42.0	31.8	28.7		
	2%	DB	35.8	42.6	51.3	61.0	65.3	67.5	70.3	71.4	67.2	55.2	46.9	39.3		
		MCWB	23.3	27.7	32.0	38.4	44.0	49.5	53.1	52.4	49.9	39.6	29.8	25.3		
	5%	DB	31.6	38.5	47.1	56.5	61.8	65.0	67.6	68.3	62.8	51.8	43.6	35.2		
		MCWB	20.7	25.4	30.8	36.0	43.0	48.2	52.0	51.4	45.9	36.5	28.6	23.0		
	10%	DB	27.6	34.1	43.0	52.4	58.4	62.3	64.5	65.1	58.6	48.7	40.0	31.0		
		MCWB	18.3	22.8	29.1	34.4	41.5	47.0	51.6	50.3	44.6	35.2	26.7	20.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	30.2	36.2	43.2	49.6	54.9	57.7	59.2	57.9	44.1	33.0	28.5		
		MCDB	40.3	45.3	49.7	61.1	62.0	65.4	67.0	71.1	69.6	57.2	48.0	44.7		
	2%	WB	23.6	27.8	33.5	39.7	47.1	52.7	55.6	56.4	53.3	41.3	30.7	25.7		
		MCDB	35.1	42.0	46.2	56.4	59.9	62.1	65.1	65.6	62.7	51.3	43.7	38.6		
	5%	WB	21.0	25.4	31.6	37.8	45.2	51.0	54.2	54.1	50.0	38.9	29.2	22.9		
		MCDB	31.1	37.4	44.1	52.9	56.4	59.9	63.1	62.9	58.0	47.8	42.2	34.6		
	10%	WB	18.4	23.1	29.6	35.6	43.6	49.3	52.7	52.3	47.4	37.1	27.1	20.4		
		MCDB	27.1	33.5	41.3	49.9	54.2	58.1	60.9	60.2	54.6	45.2	39.5	30.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	34.7	32.2	28.8	26.1	22.3	20.3	20.2	23.2	24.1	26.0	35.1	37.0		
		MCDBR	40.9	38.5	36.1	35.6	30.2	26.7	29.2	30.7	33.3	31.9	39.3	42.1		
	5% WB	MCWBR	30.1	25.5	21.1	17.1	13.4	11.5	13.3	13.6	15.5	16.9	24.0	29.8		
		MCWBR	39.0	35.6	30.4	29.6	23.4	20.5	21.2	21.4	22.6	25.4	36.3	40.7		
(40) Clear-Sky Solar Irradiance	taub		0.218	0.252	0.303	0.341	0.346	0.337	0.325	0.319	0.296	0.267	0.229	0.208		
		taud	2.542	2.445	2.275	2.186	2.274	2.399	2.485	2.496	2.538	2.549	2.568	2.593		
	Ebn at Noon		324	319	307	298	297	298	301	302	307	310	316	323		
		Edh at Noon	24	30	40	46	43	38	34	33	30	27	23	22		
(44) All-Sky Solar Radiation	RadAvg		1059	1299	1655	1911	1899	1821	1859	1770	1465	1334	1155	1012		
	RadStd		66	63	80	94	97	148	183	139	119	89	46	31		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	-205	+63	N/A

Nomenclature: See separate page